Valid Time Integration for PyElastica

Two-Level Stepping, Stability Constraints, and CPG Synchronization

Snake-HRL Project — Reference Document

1 Overview

PyElastica simulates Cosserat rod dynamics using explicit symplectic integrators. In the Snake-HRL project, the simulation is embedded inside a reinforcement learning (RL) loop, producing a *two-level time-stepping hierarchy*: a fast inner loop that integrates the rod mechanics, and a slower outer loop at which the RL agent selects actions. A *valid* time integration is one in which:

1. The inner substep Δt_{sub} is small enough for the symplectic integrator to remain stable given the rod's material stiffness and geometry.
2. The CPG (Central Pattern Generator) curvature field is updated at every substep, not just at RL boundaries, so the actuation is temporally smooth.
3. Friction forces are applied before each integration substep so they participate in the force balance.
4. Numerical damping is low enough to preserve locomotion dynamics but high enough to suppress spurious high-frequency modes.

2 Cosserat Rod Equations of Motion

Let $\mathbf{r}(s,t) \in \mathbb{R}^3$ denote the centerline position and $\mathbf{Q}(s,t) \in SO(3)$ the material frame of a Cosserat rod parameterized by arc length $s \in [0, L]$. The governing PDEs are:

$$\rho A\, \ddot{\mathbf{r}} = \frac{\partial \mathbf{n}}{\partial s} + \mathbf{f}_{\text{ext}}, \tag{1}$$

$$\rho \mathbf{I}\, \dot{\boldsymbol{\omega}} = \frac{\partial \mathbf{m}}{\partial s} + \frac{\partial \mathbf{r}}{\partial s} \times \mathbf{n} + \boldsymbol{\tau}_{\text{ext}}, \tag{2}$$

where $\mathbf{n}$ is the internal force resultant, $\mathbf{m}$ the internal moment, ρ the density, A the cross-sectional area, $\mathbf{I}$ the second moment of area tensor, $\boldsymbol{\omega}$ the angular velocity, and $\mathbf{f}_{\text{ext}}$, $\boldsymbol{\tau}_{\text{ext}}$ are external forces and torques (gravity, friction).

The constitutive relations for an isotropic rod are:

$$\mathbf{m} = \mathbf{B}\,(\boldsymbol{\kappa} - \boldsymbol{\kappa}_{\text{rest}}), \tag{3}$$

$$\mathbf{n} = \mathbf{S}\,(\boldsymbol{\sigma} - \boldsymbol{\sigma}_0), \tag{4}$$

where $\mathbf{B} = \mathrm{diag}(EI_1, EI_2, GJ)$ is the bending/torsion stiffness matrix, $\mathbf{S}$ the shear/stretch stiffness, $\boldsymbol{\kappa}$ the curvature strain vector, and $\boldsymbol{\kappa}_{\mathrm{rest}}$ the *rest curvature* that serves as the control input.

The bending energy that drives actuation is:

$$E_{\mathrm{bend}} = \int_0^L \frac{1}{2} \left(\boldsymbol{\kappa} - \boldsymbol{\kappa}_{\mathrm{rest}}\right)^\top \mathbf{B} \left(\boldsymbol{\kappa} - \boldsymbol{\kappa}_{\mathrm{rest}}\right) ds. \tag{5}$$

Setting $\boldsymbol{\kappa}_{\mathrm{rest}}$ to a time-varying serpenoid profile makes the elastic restoring forces produce undulatory locomotion.

3 Two-Level Time-Stepping Hierarchy

Table 1: Time-stepping levels and their default values.

Level	**Symbol**	**Default**	**Description**
Integration substep	Δt_{sub}	0.001 s	Smallest timestep passed to the symplectic integrator
Intermediate block	Δt	0.05 s	$N_{\mathrm{sub}} = 50$ substeps; legacy config unit
RL action step	Δt_{RL}	0.5 s	$N_{\mathrm{RL}} = 500$ substeps per agent decision

The relationships are:

$$\Delta t_{\mathrm{sub}} = \frac{\Delta t}{N_{\mathrm{sub}}}, \qquad \Delta t_{\mathrm{RL}} = N_{\mathrm{RL}} \cdot \Delta t_{\mathrm{sub}}. \tag{6}$$

At each RL step the agent outputs action parameters (amplitude, frequency, wave number, phase, turn bias). These parameters are held fixed for N_{RL} substeps, but the resulting curvature field evolves continuously because the serpenoid curve depends on the running time t.

4 Symplectic Integrators

PyElastica provides two symplectic integrators:

4.1 Position Verlet (Default)

The Störmer–Verlet method is a second-order symplectic integrator. For a system $\dot{\mathbf{q}} = \mathbf{v}$, $\dot{\mathbf{v}} = \mathbf{a}(\mathbf{q})$, one full step from t to $t + \Delta t_{\text{sub}}$ is:

$$\mathbf{q}_{1/2} \leftarrow \mathbf{q}_n + \tfrac{1}{2}\Delta t_{\text{sub}}\, \mathbf{v}_n, \tag{7}$$

$$\mathbf{v}_{n+1} \leftarrow \mathbf{v}_n + \Delta t_{\text{sub}}\, \mathbf{a}(\mathbf{q}_{1/2}), \tag{8}$$

$$\mathbf{q}_{n+1} \leftarrow \mathbf{q}_{1/2} + \tfrac{1}{2}\Delta t_{\text{sub}}\, \mathbf{v}_{n+1}. \tag{9}$$

This is time-reversible and preserves the symplectic structure, meaning energy drift is bounded over long integrations.

4.2 PEFRL (Position Extended Forest–Ruth–Like)

A fourth-order symplectic integrator with optimized coefficients [3]. It uses four force evaluations per step (versus one for Verlet) but achieves much smaller energy error at the same Δt_{sub}. The update sequence is:

$$\begin{aligned}
\mathbf{q}_1 &\leftarrow \mathbf{q}_n + \xi\, \Delta t_{\text{sub}}\, \mathbf{v}_n, \\
\mathbf{v}_1 &\leftarrow \mathbf{v}_n + (1-2\lambda)\tfrac{\Delta t_{\text{sub}}}{2}\, \mathbf{a}(\mathbf{q}_1), \\
\mathbf{q}_2 &\leftarrow \mathbf{q}_1 + \chi\, \Delta t_{\text{sub}}\, \mathbf{v}_1, \\
\mathbf{v}_2 &\leftarrow \mathbf{v}_1 + \lambda\, \Delta t_{\text{sub}}\, \mathbf{a}(\mathbf{q}_2), \\
\mathbf{q}_3 &\leftarrow \mathbf{q}_2 + (1-2(\chi+\xi))\, \Delta t_{\text{sub}}\, \mathbf{v}_2, \\
\mathbf{v}_3 &\leftarrow \mathbf{v}_2 + \lambda\, \Delta t_{\text{sub}}\, \mathbf{a}(\mathbf{q}_3), \\
\mathbf{q}_4 &\leftarrow \mathbf{q}_3 + \chi\, \Delta t_{\text{sub}}\, \mathbf{v}_3, \\
\mathbf{v}_{n+1} &\leftarrow \mathbf{v}_3 + (1-2\lambda)\tfrac{\Delta t_{\text{sub}}}{2}\, \mathbf{a}(\mathbf{q}_4), \\
\mathbf{q}_{n+1} &\leftarrow \mathbf{q}_4 + \xi\, \Delta t_{\text{sub}}\, \mathbf{v}_{n+1},
\end{aligned} \tag{10}$$

with $\xi \approx 0.1786$, $\lambda \approx -0.2123$, $\chi \approx -0.6626$.

5 Stability Constraints

Since both integrators are explicit, the substep Δt_{sub} must satisfy a CFL-like condition dictated by the stiffest mode in the discretized rod. The critical timescale is set by bending wave propagation:

$$\Delta t_{\text{sub}} < C\sqrt{\frac{\rho A\, \Delta s^4}{EI}}, \tag{11}$$

where $\Delta s = L/N_e$ is the element length, N_e the number of elements, and C is an order-unity constant that depends on the integrator (smaller for Verlet, larger for PEFRL).

Table 2: Default material and geometry parameters.

Parameter	**Symbol**	**Value**
Young's modulus	E	10^5 Pa
Poisson ratio	ν	0.5
Shear modulus	$G = E/2(1+\nu)$	3.33×10^4 Pa
Density	ρ	1200 kg/m^3
Rod length	L	0.5 m
Rod radius	r	0.02 m
Number of elements	N_e	20
Element length	Δs	0.025 m
Second moment of area	$I = \pi r^4/4$	1.257×10^{-7} m^4

Substituting the default parameters into (11):

$$\sqrt{\frac{\rho A\,\Delta s^4}{EI}} = \sqrt{\frac{1200 \times \pi (0.02)^2 \times (0.025)^4}{10^5 \times 1.257 \times 10^{-7}}} \approx 1.7 \times 10^{-3}\ \text{s}. \tag{12}$$

The default substep of $\Delta t_{\text{sub}} = 10^{-3}$ s sits safely below this limit, providing a stability margin of roughly $1.7\times$.

5.1 Numerical Damping

PyElastica applies an `AnalyticalLinearDamper` with coefficient γ. This adds a dissipative term $-\gamma \mathbf{v}$ to each node's velocity at every substep.

- $\gamma = 0.002$: default for locomotion — preserves undulatory dynamics while suppressing numerical ringing.
- $\gamma \geq 0.008$: **freezes the rod** completely; overdamped regime where elastic restoring forces cannot overcome dissipation.

The valid range is therefore $\gamma \in (0,\ 0.008)$.

6 CPG–Integrator Synchronization

The serpenoid curvature profile is:

$$\kappa(s_j, t) = A \sin\!\big(k\, s_j + \omega\, t + \phi\big) + \kappa_{\text{turn}}, \tag{13}$$

where s_j are joint arc-length positions, A is amplitude, k the spatial wave number, $\omega = 2\pi f$ the temporal frequency, ϕ a phase offset, and κ_{turn} a constant steering bias.

A valid integration requires that $\boldsymbol{\kappa}_{\text{rest}}$ in (3) is updated from (13) *at every substep*, not just at RL decision boundaries. Between RL steps, the action parameters $(A, f, k, \phi, \kappa_{\text{turn}})$ are held constant but t advances by Δt_{sub} each substep, so the curvature field evolves smoothly.

7 Ground Friction at Each Substep

Anisotropic Resistive Force Theory (RFT) friction is applied before each integration substep:

$$\mathbf{f}_{\text{RFT}} = -c_t\,\mathbf{v}_t - c_n\,\mathbf{v}_n, \tag{14}$$

where $\mathbf{v}_t$ and $\mathbf{v}_n$ are the tangential and normal components of node velocity, with default coefficients $c_t = 0.01$ and $c_n = 0.05$ (a 5:1 anisotropy ratio enabling serpentine propulsion).

The friction forces must be included in the external forcing term $\mathbf{f}_{\text{ext}}$ of (1) at every substep. Applying them only at RL boundaries would produce non-physical jerky dynamics and destabilize the integrator.

8 Contrast: DisMech Implicit Time Integration

The project supports a second physics backend, DisMech, which uses *implicit* (backward Euler) time integration. Understanding the differences clarifies why the validity conditions for Elastica exist.

8.1 Backward Euler Formulation

DisMech discretizes the equations of motion (1)–(2) using the backward Euler scheme. Given the state vector $\mathbf{x} = [\mathbf{q}^\top,\, \mathbf{v}^\top]^\top$ (positions and velocities of all N nodes), the update rule is:

$$\mathbf{v}_{n+1} = \mathbf{v}_n + \Delta t\,\mathbf{M}^{-1}\mathbf{F}(\mathbf{q}_{n+1},\, \mathbf{v}_{n+1}), \tag{15}$$

$$\mathbf{q}_{n+1} = \mathbf{q}_n + \Delta t\,\mathbf{v}_{n+1}, \tag{16}$$

where $\mathbf{M}$ is the mass matrix and $\mathbf{F}$ includes all internal elastic forces, damping, and external forces. Crucially, $\mathbf{F}$ is evaluated at the *unknown* future state $(\mathbf{q}_{n+1}, \mathbf{v}_{n+1})$, making the system implicit.

Substituting (16) into (15) and rearranging gives the nonlinear residual:

$$\mathbf{g}(\mathbf{q}_{n+1}) \coloneqq \mathbf{M}\big(\mathbf{q}_{n+1} - 2\mathbf{q}_n + \mathbf{q}_{n-1}\big) - \Delta t^2\,\mathbf{F}(\mathbf{q}_{n+1}) = \mathbf{0}, \tag{17}$$

which is solved via Newton's method at each timestep.

8.2 Newton Solver

Each timestep requires iterating:

$$\mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} - \mathbf{J}^{-1}(\mathbf{q}^{(k)})\,\mathbf{g}(\mathbf{q}^{(k)}), \tag{18}$$

where $\mathbf{J} = \partial\mathbf{g}/\partial\mathbf{q}$ is the Jacobian (tangent stiffness matrix plus mass and damping contributions). Convergence is declared when:

$$\|\mathbf{g}(\mathbf{q}^{(k)})\| < \epsilon_f \quad \text{or} \quad \|\mathbf{q}^{(k+1)} - \mathbf{q}^{(k)}\| < \epsilon_d, \tag{19}$$

with default tolerances $\epsilon_f = 10^{-4}$ (force) and $\epsilon_d = 10^{-2}$ (displacement), and a maximum of 25 iterations.

8.3 Key Differences from PyElastica

Table 3: Comparison of time integration approaches.

Property	**PyElastica (Explicit)**	**DisMech (Implicit)**
Integration scheme	Symplectic (Verlet / PEFRL)	Backward Euler (Newton)
Timestep per call	$\Delta t_{\text{sub}} = 0.001$ s	$\Delta t = 0.05$ s
Steps per second	1000	20
Stability	*Conditional* ($\Delta t_{\text{sub}} < \Delta t_{\text{sub}}^{\max}$)	*Unconditional*
Substep loop	$N_{\text{RL}} = 500$ substeps per RL action	1 step per RL action
Cost per step	Low (explicit evaluation)	High (Jacobian assembly + linear solve)
Energy behavior	Bounded drift (symplectic)	Numerical dissipation (implicit)
Curvature control	`rest_kappa` array	Natural strain of bend springs
Solver parameters	Δt_{sub}, γ (damping)	ϵ_f, ϵ_d, max_iter

The fundamental trade-off is:

- **PyElastica** takes many cheap explicit steps. Each substep is a simple algebraic update with no matrix inversion, but the timestep is constrained by the CFL condition (11). The symplectic structure ensures long-term energy conservation.
- **DisMech** takes few expensive implicit steps. Each step requires assembling and factorizing the Jacobian $\mathbf{J}$ (typically sparse, solved via PARDISO in the C++ backend), but the method is unconditionally stable so Δt can be $50\times$ larger. However, backward Euler introduces *numerical dissipation* that damps high-frequency modes — acceptable for locomotion but less faithful for energy-sensitive analyses.

Algorithm 1 Single RL Step with PyElastica Integration

Require: Action $\mathbf{a} = (A,\ f,\ k,\ \phi,\ \kappa_{\text{turn}})$
Require: Substep size Δt_{sub}, substeps per action N_{RL}
Require: Rod state $(\mathbf{q},\ \mathbf{v})$, simulation time t
Ensure: Updated state $(\mathbf{q}',\ \mathbf{v}')$, new time t'

```
ω ← 2πf                                          ▷ Angular frequency
for i = 1 to N_RL do
    for j = 1 to N_e − 1 do                      ▷ Update rest curvature
        κ_rest,j ← A sin(k s_j + ω t + φ) + κ_turn
    end for
    f_ext ← ComputeRFT(v, c_t, c_n)              ▷ Friction
    (q, v, t) ← SymplecticStep(q, v, t, Δt_sub)
end for
return (q, v, t)
```

8.4 Why Validity Conditions Differ

Because DisMech is unconditionally stable, many of the Elastica validity conditions in Section 10 do not apply:

1. **No CFL constraint** — DisMech can use $\Delta t = 0.05$ s directly, without substeps.
2. **No per-substep curvature update** — with only one step per RL action, curvature is set once at the step boundary (the CPG profile is evaluated at t_n, not continuously during the step).
3. **No explicit damping coefficient** — numerical dissipation from backward Euler itself provides damping; an additional damping viscosity is optional (default 0.0).
4. **Convergence monitoring replaces stability monitoring** — the relevant diagnostic is the Newton residual norm $\|\mathbf{g}\|$, not the energy drift.

This means DisMech is simpler to configure but offers less temporal resolution of the CPG waveform within each RL step. The continuous curvature evolution that Elastica captures across 500 substeps is collapsed to a single snapshot in DisMech.

9 Algorithms

10 Summary of Validity Conditions

A time integration is *valid* if and only if all conditions for the chosen backend are satisfied.

10.1 PyElastica (Explicit Symplectic)

1. **Substep stability:** $\Delta t_{\text{sub}} < C\sqrt{\rho A\, \Delta s^4/(EI)}$, satisfied by $\Delta t_{\text{sub}} = 10^{-3}$ s with the default parameters.

2. **Symplectic integrator:** Either Position Verlet (second-order) or PEFRL (fourth-order)

Algorithm 2 Position Verlet Substep (SYMPLECTICSTEP)

Require: Positions $\mathbf{q}$, velocities $\mathbf{v}$, time t, substep Δt_{sub}
Ensure: Updated $(\mathbf{q}', \mathbf{v}', t')$

$\mathbf{q}_{1/2} \leftarrow \mathbf{q} + \frac{1}{2}\Delta t_{\text{sub}}\,\mathbf{v}$ ▷ Half-step position
$\mathbf{a} \leftarrow \mathbf{a}(\mathbf{q}_{1/2})$ ▷ Acceleration from internal + external forces
$\mathbf{v}' \leftarrow \mathbf{v} + \Delta t_{\text{sub}}\,\mathbf{a}$ ▷ Full-step velocity
$\mathbf{q}' \leftarrow \mathbf{q}_{1/2} + \frac{1}{2}\Delta t_{\text{sub}}\,\mathbf{v}'$ ▷ Half-step position
$\mathbf{v}' \leftarrow \mathbf{v}' - \gamma\,\mathbf{v}'$ ▷ Damping ($\gamma \ll 1$)
$t' \leftarrow t + \Delta t_{\text{sub}}$
return $(\mathbf{q}', \mathbf{v}', t')$

Algorithm 3 Resistive Force Theory (COMPUTERFT)

Require: Node velocities $\mathbf{v} \in \mathbb{R}^{3\times N}$, drag coefficients c_t, c_n
Ensure: External friction forces $\mathbf{f} \in \mathbb{R}^{3\times N}$

for $i = 1$ **to** N **do**
  $\hat{\mathbf{t}}_i \leftarrow$ unit tangent at node i ▷ From adjacent positions
  $v_{t,i} \leftarrow (\mathbf{v}_i \cdot \hat{\mathbf{t}}_i)\,\hat{\mathbf{t}}_i$ ▷ Tangential component
  $v_{n,i} \leftarrow \mathbf{v}_i - v_{t,i}$ ▷ Normal component
  $\mathbf{f}_i \leftarrow -c_t\,v_{t,i} - c_n\,v_{n,i}$
end for
return $\mathbf{f}$

is used. Non-symplectic integrators (e.g., forward Euler) would accumulate energy drift.

3. **Per-substep curvature update:** The serpenoid $\kappa(s_j, t)$ is recomputed at every substep using the current simulation time t, not cached from the RL boundary.

4. **Per-substep friction:** RFT (or Coulomb/Stribeck) forces are computed and applied before each integrator substep.

5. **Bounded damping:** $\gamma \in (0,\ 0.008)$ to avoid freezing the rod while suppressing numerical ringing.

6. **Consistent time accounting:** Both the integrator time and CPG phase advance by exactly Δt_{sub} per substep; no time desynchronization between actuation and physics.

10.2 DisMech (Implicit Backward Euler)

1. **Newton convergence:** The residual norm $\|\mathbf{g}\|$ must fall below ϵ_f (or displacement below ϵ_d) within the maximum iteration budget. A diverging or stagnating Newton solve indicates an invalid step.

2. **Timestep–material consistency:** While unconditionally stable, excessively large Δt degrades accuracy through numerical dissipation. The default $\Delta t = 0.05$ s is validated against the locomotion frequency band.

Algorithm 4 Single RL Step with DisMech Integration (Implicit)

Require: Action $\mathbf{a} = (A,\ f,\ k,\ \phi,\ \kappa_{\text{turn}})$
Require: Timestep Δt, rod state $(\mathbf{q},\ \mathbf{v})$, time t
Require: Tolerances ϵ_f, ϵ_d, max iterations K
Ensure: Updated state $(\mathbf{q}',\ \mathbf{v}')$, residual norm r

$\omega \leftarrow 2\pi f$
for $j = 1$ **to** $N_e - 1$ **do** ▷ Set curvature once per RL step
  $\kappa_{\text{nat},j} \leftarrow A \sin(k\, s_j + \omega\, t + \phi) + \kappa_{\text{turn}}$
end for
$\mathbf{q}^{(0)} \leftarrow \mathbf{q} + \Delta t\, \mathbf{v}$ ▷ Initial guess (explicit Euler)
for $k = 0$ **to** $K - 1$ **do** ▷ Newton iterations
  $\mathbf{g} \leftarrow \mathbf{M}(\mathbf{q}^{(k)} - 2\mathbf{q} + \mathbf{q}_{\text{prev}}) - \Delta t^2\, \mathbf{F}(\mathbf{q}^{(k)})$
  $\mathbf{J} \leftarrow \frac{\partial \mathbf{g}}{\partial \mathbf{q}}\big|_{\mathbf{q}^{(k)}}$ ▷ Jacobian (sparse)
  $\delta\mathbf{q} \leftarrow \mathbf{J}^{-1}\mathbf{g}$ ▷ Sparse linear solve
  $\mathbf{q}^{(k+1)} \leftarrow \mathbf{q}^{(k)} - \delta\mathbf{q}$
  if $\|\mathbf{g}\| < \epsilon_f$ **or** $\|\delta\mathbf{q}\| < \epsilon_d$ **then**
    break ▷ Converged
  end if
end for
$\mathbf{v}' \leftarrow (\mathbf{q}^{(k+1)} - \mathbf{q})/\Delta t$
$r \leftarrow \|\mathbf{g}\|$ ▷ Final residual for diagnostics
return $(\mathbf{q}^{(k+1)},\ \mathbf{v}',\ r)$

3. **Curvature set before solve:** The natural strain of each bend spring must be set to the target curvature *before* calling the Newton solver, so the elastic energy landscape reflects the desired actuation.

4. **Residual monitoring:** The returned residual norm should be logged; values above ϵ_f indicate incomplete convergence and unreliable state.

References

[1] M. Gazzola, L. H. Dudte, A. G. McCormick, and L. Mahadevan, "Forward and inverse problems in the mechanics of soft filaments," *Royal Society Open Science*, vol. 5, no. 6, 2018.

[2] X. Zhang, F. K. Chan, T. Parthasarathy, and M. Gazzola, "Modeling and simulation of complex dynamic musculoskeletal architectures," *Nature Communications*, vol. 10, 2019.

[3] I. P. Omelyan, I. M. Mryglod, and R. Folk, "Optimized Forest–Ruth- and Suzuki-like algorithms for integration of motion in many-body systems," *Computer Physics Communications*, vol. 146, pp. 188–199, 2002.

[4] M. Bergou, M. Wardetzky, S. Robinson, B. Audoly, and E. Grinspun, "Discrete elastic rods," *ACM Transactions on Graphics (SIGGRAPH)*, vol. 27, no. 3, 2008.

[5] M. Bergou, B. Audoly, E. Vouga, M. Wardetzky, and E. Grinspun, "Discrete viscous threads," *ACM Transactions on Graphics (SIGGRAPH)*, vol. 29, no. 4, 2010.